Deepak Mallampati

FORWARD DEPLOYED ENGINEER · APPLIED AI

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US-based (F-1 OPT) · open to Zurich relocation

PROFESSIONAL SUMMARY

Applied AI / Forward Deployed Engineer with 4+ years building and deploying production AI on messy real-world data — from operational/industrial sensor data to regulated enterprise corpora. Embeds with stakeholders to integrate data and workflows, ships features from field feedback, and applies state-of-the-art LLM, RAG, and ML in challenging settings. Track record of measurable impact: 42% lower token cost, 92% retrieval precision, 60% fewer failed production runs, and p95 latency cut from 1.5s to 800ms.

CORE COMPETENCIES

Forward Deployed / Customer-Embedded Delivery • Applied AI & LLM Systems • RAG & Vector Retrieval • Python • Anomaly Detection & Failure Prediction • Data Integration & Pipelines • MLOps (Docker, Kubernetes, CI/CD) • SQL

Generative AI & LLMs: GPT-4o/5, Claude, Gemini, LLaMA | LangChain/LangGraph | RAG | Multi-Agent Systems | Fine-tuning (LoRA/QLoRA)

ML & Data: PyTorch, TensorFlow | Anomaly Detection, Forecasting, Classification | NER | Drift Detection | RAGAS, BERTScore | Pandas, NumPy, Spark

Deployment & Infra: AWS (Bedrock, SageMaker, Lambda, EC2, S3), Azure, GCP | Docker, Kubernetes, Terraform | CI/CD | MLflow, W&B | FastAPI

Data Stores: PostgreSQL, MongoDB, Redis | FAISS, Pinecone, Weaviate

PROFESSIONAL EXPERIENCE

AI Engineer — Progress Solutions *Jul 2025 – Present | USA*

Applied AI · agentic systems, RAG, and reliable production delivery

- Architected production **agentic LLM systems** on a conductor–subagent framework, cutting token consumption **42%** while sustaining accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large document corpora (dense retrieval + cross-encoder re-ranking) to lift answer relevance and reduce hallucination on domain-specific queries.
- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retries, reducing failed production runs by **60%**.
- Stood up an **LLM evaluation & monitoring** framework (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to catch regressions pre-deploy.

AI Engineer Intern — Vanguard *Jan 2024 – Jan 2025 | USA*

Internal AI platform · delivery across 20+ teams

- Built and enhanced **AI agents and backend services** for an internal AI platform spanning **25+ agents**, integrating with production pipelines and observability tooling.
- Evaluated GPT-4o, Claude, and Gemini and tuned prompts, improving task success by **27%** and informing platform model selection.
- Shipped **12 APIs/microservices**, cutting p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ teams.

ML Trainee — Emerson *Jun 2022 – Apr 2023 | India*

Industrial / operational ML · equipment-failure prediction

- Built supervised **regression and classification models** on operational pipeline-storage data for equipment-failure prediction, anomaly detection, and capacity forecasting.
- Engineered features and ran end-to-end preprocessing of raw **sensor and operational datasets** into model-ready pipelines.
- Fine-tuned BERT/RoBERTa for entity extraction and classification, achieving **89% F1** on custom NER tasks.

ML & GenAI Research Assistant — Kent State University *Oct 2023 – Jan 2024 | USA*

Privacy-preserving ML on sensitive operational data

- Built a **privacy-preserving ML pipeline** (k-anonymity, l-diversity, Laplace differential privacy), cutting re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.
- Designed a composite privacy–utility scoring framework to select the optimal anonymization operating point.

PROJECTS

career-ops

Autonomous AI job-search pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision — a deployed, real-world automation system.

MangaLens

Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites — production AI in the wild.

Privacy-Preserving Clinical ML Pipeline

Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive data without exposing identities; quantifies the privacy-utility tradeoff.

EDUCATION

M.S. in Computer Science — *Kent State University, OH* *2025 | GPA 3.70/4.0*